

Concepts:

- What are Agile and DevOps?
- What is the meaning of on-premise service?
- What is the meaning of airgapped service?

Network:

- What happens when I run `https://google.com`?
- How do packets transmit across the internet? (OSI)
- What are symmetric, asymmetric encryption?
- What is DNS?
- What is CDN?
- How does TLS work?

Docker:

- VM vs Container?
- What is Docker?
- What is Docker Architecture?
- What are Volumes?
- What are best practices for a Dockerfile? (layer caching, non Root, multistage, base image)
- What is Docker Compose?

Kubernetes:

- What are the main Kubernetes components? (etcd, scheduler, api server, kube-proxy, coredns)
- What are the main Kubernetes objects? (deployment, service, job, cronjob, svc, statefulset, daemonset, ingress, RBAC)

- Deployment Patterns? (blue-green/rolling update/canary/A-B)
- Horizontal vs Vertical Scaling?
- What are Operators and CRDs?
- What is Service-Mesh?
- What is CNI? (name some)
- Name some security features to have in a cluster. (4C => cloud, cluster, container, code)

Automation:

- What is IaC?
- What is GitOps?
- What is CI/CD?
- What is ArgoCD?
- What is Gitlab-CI?
- What is Ansible?
- What is Terraform?
- What is Bash Script?

Monitoring:

- What is Log? (Elastic-Fluentbit-Kibana-Loki-Alloy)
- What is Metric? (Prometheus-VM-Grafana-HA solutions(thanos-mimir))
- How does prometheus work? (exporters, scraping, metric types(gauge, counter))
- What is Trace? (Jaeger-Tempo)

Linux:

- What are linux permissions? (user, group, other, root, XYZ(740, 600))

- What are ssh, scp, jump server?
- How to add a disk to the server? (partitioning, formatting, mounting)
- What is the “too many open files” error?
- What is the /etc/resolv.conf folder?
- What is Firewall?

Scenarios:

- I am a big company with trouble using your service as a third party company. What do you do to solve the issue? (problem solving power, networking issue, bandwidth checking, resource usage checking)
- We have a Postgres cluster which is slow or even down. What should we do as a devops engineer? (HA, Pod Anti Affinity, LB)
- In my k8s cluster i always get connection refused. What can be the problem? (dns, improper destination host setting)
- Tell me some disaster plans for the company? (db backup, retention, Automation, temporary fallback plan for down time)
- What was a disaster/incident that you have noticed and solved?
- What happens when my container restarts and its metrics are set to 0? How should we handle this bias in our monitoring system like grafana? (using increase, rate, ... functions to measure relative and not absolute)
- A rocket has hit the data center. What is your plan to recover the company? What is your plan to prevent problems from happening again?